What is Switching?

Switching is the process of transferring data packets from one device to another in a network, or from one network to another, using specific devices called **switches**. A computer user experiences switching all the time for example, accessing the Internet from your computer device, whenever a user requests a webpage to open, the request is processed through switching of data packets only.

Switching takes place at the Data Link layer of the OSI Model. This means that after the generation of data packets in the Physical Layer, switching is the immediate next process in data communication.

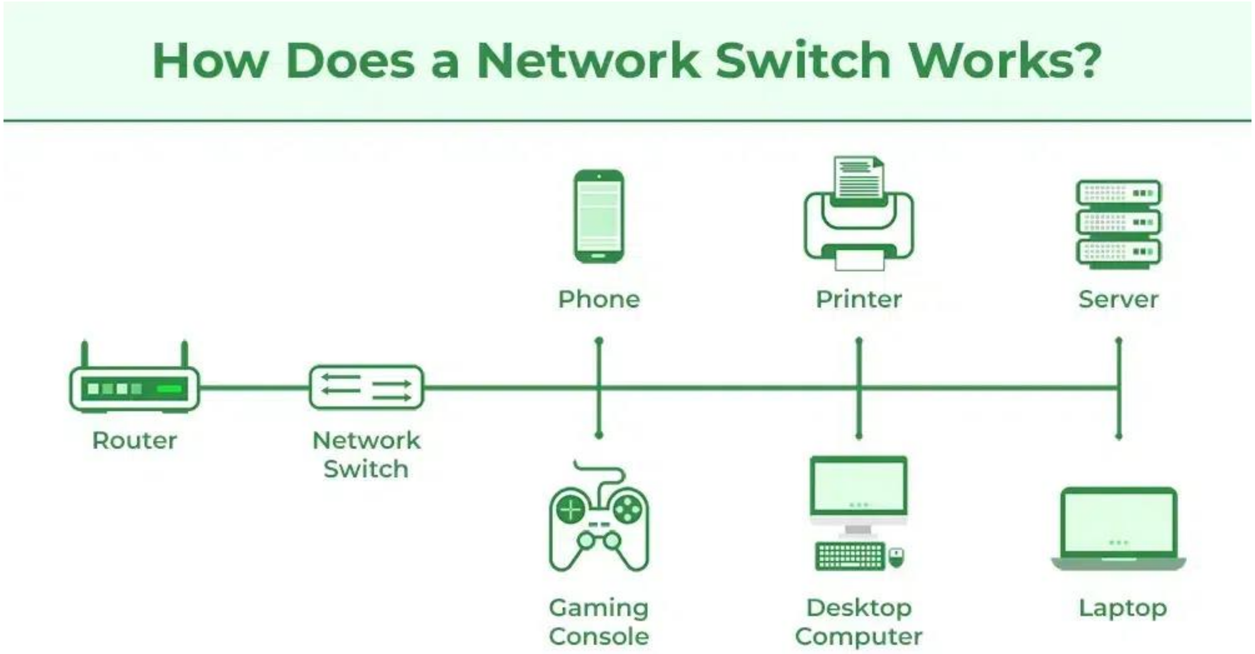
Introduction to Switch

- A switch is a hardware device in a network that connects and helps multiple devices share a network without their data interfering with each other.
- A switch works like a traffic cop at a busy intersection. When a data packet arrives, the switch decides where it needs to go and sends it through the right port.
- Some data packets come from devices directly connected to the switch, like computers or VoIP phones. Other packets come from devices connected through hubs or routers.
- The switch knows which devices are connected to it and can send data directly between them. If the data needs to go to another network, the switch sends it to a router, which forwards it to the correct destination.

What is Network Switching?

A switch is a dedicated piece of computer hardware that facilitates the process of switching i.e., incoming data packets and transferring them to their destination. A switch works at the Data Link layer of the OSI Model. A switch primarily handles the incoming data packets from a source computer or network and decides the appropriate port through which the data packets will reach their target computer or network.

A switch decides the port through which a data packet shall pass with the help of its destination MAC(Media Access Control) Address. A switch does this effectively by maintaining a switching table, (also known as forwarding table). A network switch is more efficient than a network Hub or repeater because it maintains a switching table, which simplifies its task and reduces congestion on a network, which effectively improves the performance of the network.



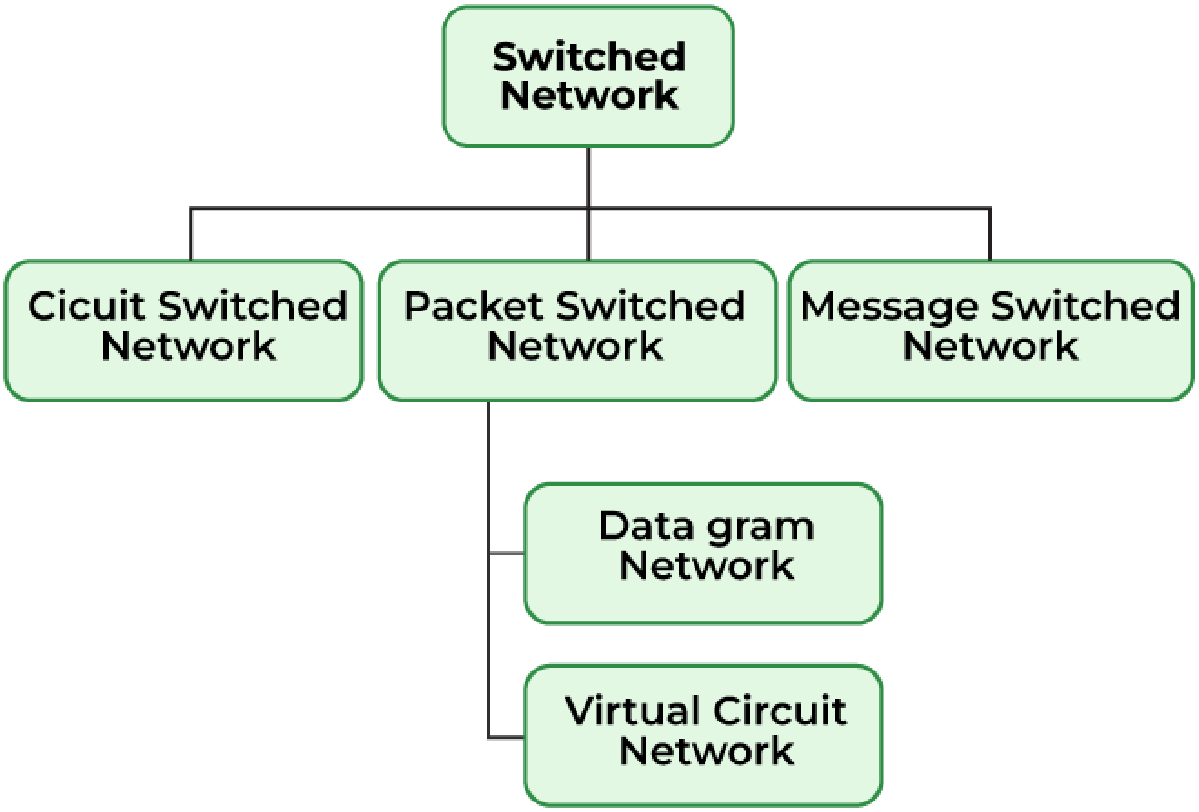
The switching process involves the following steps:

- **Frame Reception:** The switch receives a data frame or packet from a computer connected to its ports.
- **MAC Address Extraction:** The switch reads the header of the data frame and collects the destination MAC Address from it.
- **MAC Address Table Lookup:** Once the switch has retrieved the MAC Address, it performs a lookup in its Switching table to find a port that leads to the MAC Address of the data frame.
- **Forwarding Decision and Switching Table Update:** If the switch matches the destination MAC Address of the frame to the MAC address in its switching table, it forwards the data frame to the respective port. However, if the destination MAC Address does not exist in its forwarding table, it follows the flooding process, in which it sends the data frame to all its ports except the one it came from and records all the MAC Addresses to which the frame was delivered. This way, the switch finds the new MAC Address and updates its forwarding table.
- **Frame Transition:** Once the destination port is found, the switch sends the data frame to that port and forwards it to its target computer/network.

Types of Switching

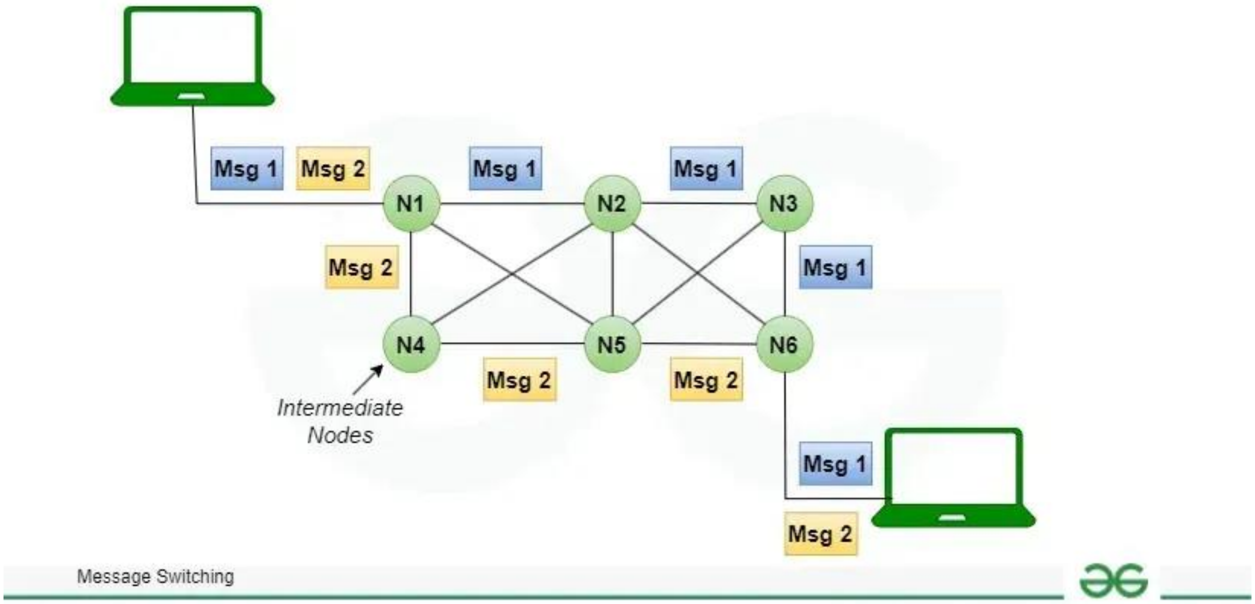
There are three types of switching methods:

- [Message Switching]
- [Circuit Switching]
- [Packet Switching]
    - Datagram Packet Switching
    - Virtual Circuit Packet Switching



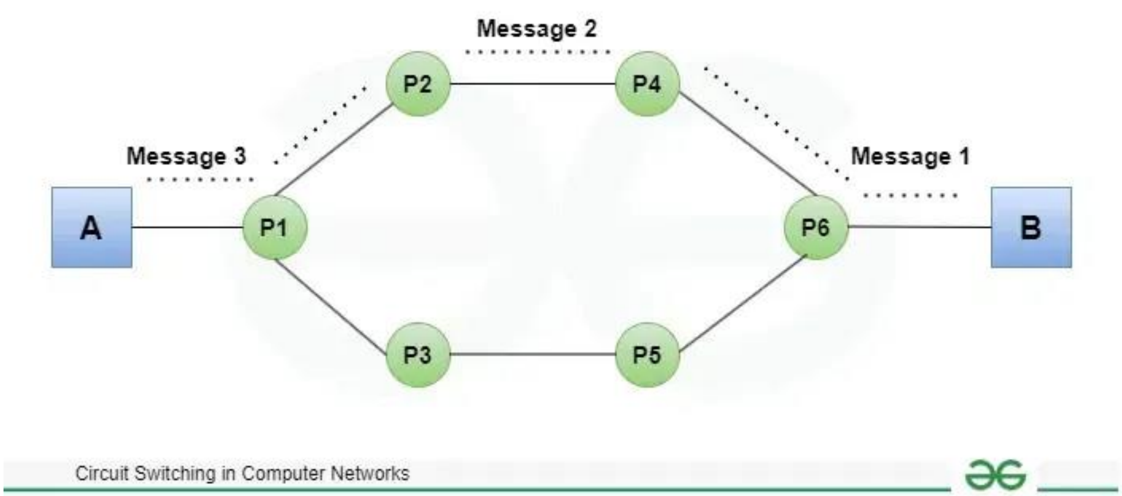
Let us now discuss them individually:

Message Switching: This is an older switching technique that has become obsolete. In message switching technique, the entire data block/message is forwarded across the entire network thus, making it highly inefficient.



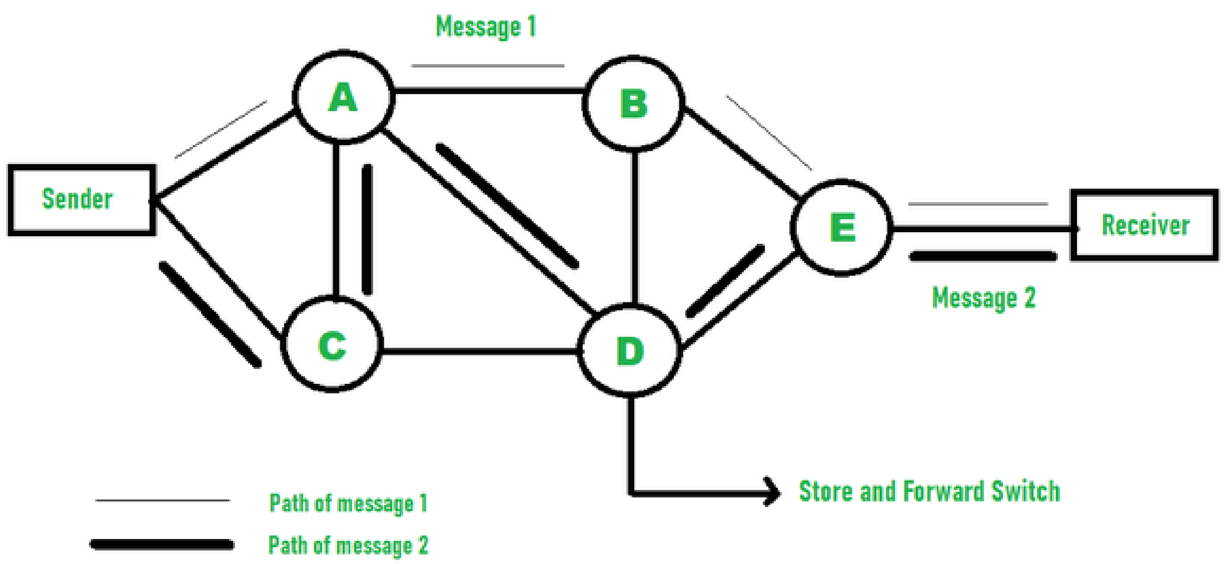
Message Switching

Circuit Switching: In this type of switching, a connection is established between the source and destination beforehand. This connection receives the complete bandwidth of the network until the data is transferred completely.
This approach is better than [message switching] as it does not involve sending data to the entire network, instead of its destination only.



Packet Switching: This technique requires the data to be broken down into smaller components, data frames, or packets. These data frames are then transferred to their destinations according to the available resources in the network at a particular time.

This switching type is used in modern computers and even the Internet. Here, each data frame contains additional information about the destination and other information required for proper transfer through network components.

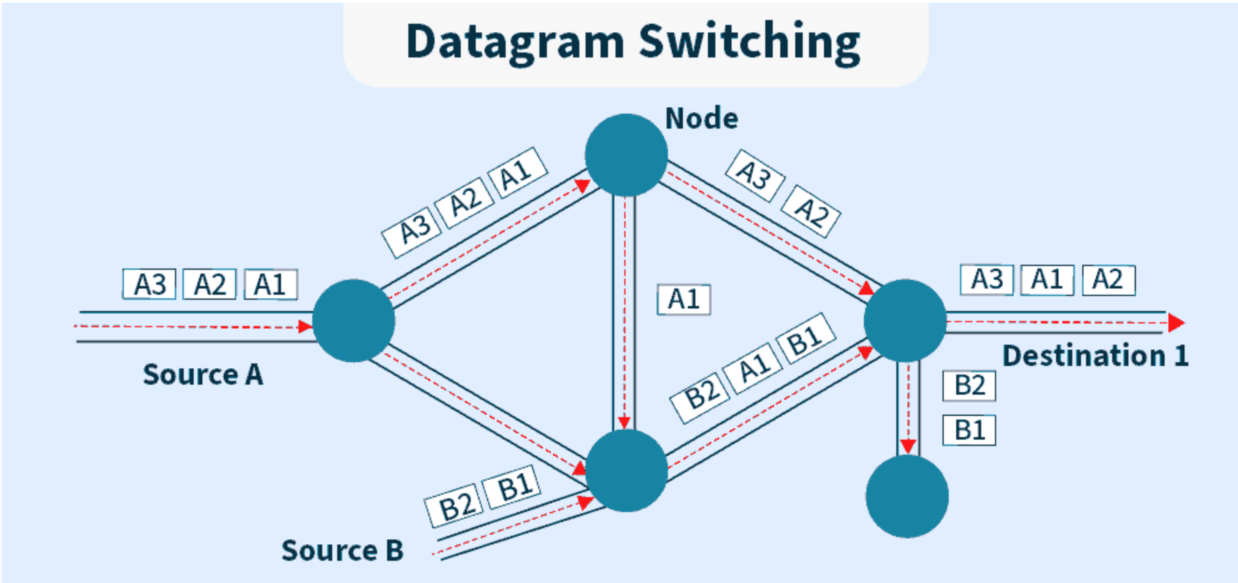


Difference between Datagram Switching and Virtual Circuit Switching

In networking, two important switching methods are Datagram Switching and Virtual Circuit Switching which are used for data transmission. These methods indicate the way data packets are transmitted in a network, and the path that packet has to follow. In this article, both techniques will be described in detail, as well as their merits compared to identify the major differences between them.

What is Datagram Switching?

Datagram switching is a packet switching method that treats each packet, or datagram, as a separate entity. Each packet is routed via the network on its own. It is a service that does not require a connection. Because there is no specific channel for a connection session, there is no need to reserve resources. As a result, packets have a header with all the destination's information. The intermediate nodes assess a packet's header and select an appropriate link to a different node closer to the destination.



Advantages of Datagram Switching

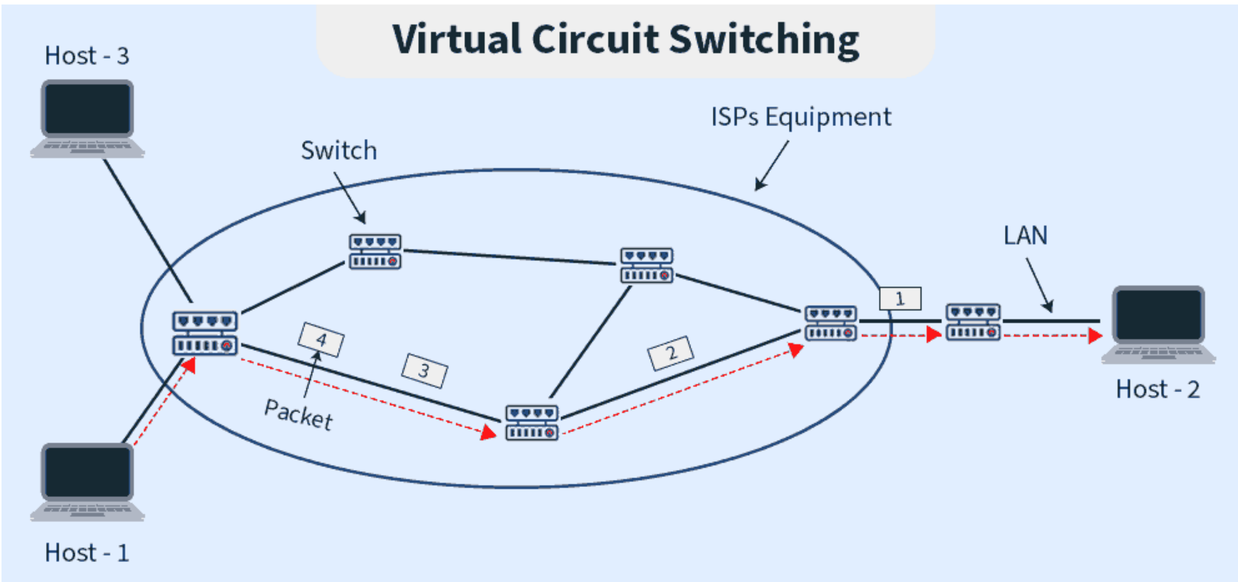
- Scalability: Datagram switching is highly scalable and can handle large amounts of traffic on a network.
- Flexibility: Datagram switching is flexible and can support variable packet sizes and data rates.
- Simple routing: Datagram switching does not require a pre-established path for each packet, allowing packets to be routed dynamically.
- Lower latency: Datagram switching typically has lower latency than [virtual circuit switching](), as packets are sent immediately without any delay for setup.

Disadvantages of Datagram Switching

- Higher error rates: Datagram switching is more susceptible to errors than virtual circuit switching, as there is no guaranteed delivery or error correction.
- Lack of QoS: Datagram switching does not provide any Quality of Service guarantees, meaning that different types of traffic may be treated equally.
- Increased network congestion: Without a pre-established path for each packet, datagram switching can lead to increased network congestion and potential delays.

What is Virtual Circuit?

Virtual packet switching approach in which a path is built between the source and the final destination through which all packets are routed throughout a call is known as virtual circuit switching. Because the connection looks to the user to be an infatuated physical circuit, this path is referred to as a virtual circuit. Other communications, on the other hand, may be sharing parts of the same path. Before the data transmission can commence, the source and destination must agree on a virtual circuit path. For the decision, all intermediary nodes between the two places add a routing entry to their routing database. Additional parameters, like the utmost packet size, also are exchanged between the source and therefore the destination during call setup. The virtual circuit is cleared after the info transfer is completed.



Advantages of Virtual Circuit Switching

- Guaranteed delivery: Virtual circuit switching provides guaranteed delivery of packets, reducing the likelihood of lost or corrupted data.
- Lower error rates: Virtual circuit switching typically has lower error rates than datagram switching due to its error correction mechanisms.
- QoS support: Virtual circuit switching supports Quality of Service guarantees, allowing for prioritization of different types of traffic.
- Efficient use of bandwidth: Virtual circuit switching establishes a pre-determined path for each packet, resulting in efficient use of bandwidth.

Disadvantages of Virtual Circuit Switching

- Limited scalability: Virtual circuit switching is less scalable than datagram switching and may not be suitable for large networks.
- Increased setup time: Virtual circuit switching requires a setup time for each connection, which can lead to increased latency and delay.

- Fixed data rates: Virtual circuit switching typically supports fixed data rates, which may not be suitable for applications that require variable packet sizes or data rates.

Some similarities between Datagram Switching and Virtual Circuit Switching

- Both are packet-switching technologies: Datagram switching and virtual circuit switching are both packet-switching technologies, which means that they break up data into small packets for transmission across a network.
- Both can handle multiple transmissions simultaneously: Both Datagram and virtual circuit switching can handle multiple transmissions simultaneously by dividing data into packets and transmitting them separately.
- Both use connection-oriented and connectionless communication: Virtual circuit switching can use both connection-oriented and connectionless communication, while Datagram switching typically uses connectionless communication. However, both techniques can support both types of communication.
- Both support error correction: Both Datagram and virtual circuit switching support error correction by using checksums or parity checks to ensure the accuracy of transmitted data.
- Both can handle variable-length packets: Datagram and virtual circuit switching can handle variable-length packets, which means that they can adjust to the size of the data being transmitted.
- Both can support QoS (Quality of Service): Both Datagram and virtual circuit switching can support QoS, which allows for the prioritization of certain types of data over others.
- Both can support fragmentation: Both Datagram and virtual circuit switching can support fragmentation, which allows for large data packets to be broken up into smaller packets for transmission across a network.

Difference between Datagram Switching and Virtual Circuit Switching

Datagram Switching	Virtual Circuit Switching
It is connection less service. There is no need for reservation of resources as there is no dedicated path for a connection session.	Virtual circuits are connection-oriented, which means that there is a reservation of resources like buffers, bandwidth, etc. for the time during

Datagram Switching	Virtual Circuit Switching
	which the new setup VC is going to be used by a data transfer session.
All packets are free to use any available path. As a result, intermediate routers calculate routes on the go due to dynamically changing routing tables on routers.	The first sent packet reserves resources at each server along the path. Subsequent packets will follow the same path as the first sent packet for the connection time.
Data packets reach the destination in random order, which means they need not reach in the order in which they were sent out.	Packets reach in order to the destination as data follows the same path.
Every packet is free to choose any path, and hence all the packets must be associated with a header containing information about the source and the upper layer data.	All the packets follow the same path and hence a global header is required only for the first packet of connection and other packets will not require it.
Datagram networks are not as reliable as Virtual Circuits.	Virtual Circuits are highly reliable.
Efficiency high, delay more	Efficiency low and delay less
But it is always easy and cost-efficient to implement datagram networks as there is no need of reserving resources and	Implementation of virtual circuits is costly as each time a new connection has to be set up with reservation of resources and extra information handling at routers.

Datagram Switching	Virtual Circuit Switching
making a dedicated path each time an application has to communicate.	
A Datagram based network is a true packet switched network. There is no fixed path for transmitting data.	A virtual circuit network uses a fixed path for a particular session, after which it breaks the connection and another path has to be set up for the next session.
Widely used in Internet	Used in X.25, ATM(Asynchronous Transfer Mode)